Tesseract - proposed civic-tech platform

A safer city starts with a single observation

Tesseract: street-level civic & road hazard tracker

“See the hazard. Pin the problem. Power the fix.”

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The everyday risk

danger is often hiding in the everyday commute



open manholes



exposed electrical wires



broken roads and footpaths



waterlogged streets

Small infrastructure failures become serious safety risks when they go unnoticed or unresolved.

Where the handoff breaks

The problem is not just the hazard. It is the gap between seeing and solving.

Reporting friction

Too many steps, forms, and unclear channels.

Poor visibility

Reports fragmented across locations, departments, and channels.

Weak accountability

Citizens can’t tell if a report was assigned, repaired, or verified.

No simple report. No unified picture. No verified closure.





The proposed response

what if every citizen could become a sensor for safer streets?

CivicLens is a proposed unified civic intelligence platform.

1. Citizen observation → photo-first report
2. Geospatial hazard map → severity-ranked queue
3. Verified repair → visible closure

one report can become actionable intelligence.

Planned features · conceptual designs

report in seconds. prioritize with clarity. verify the fix.

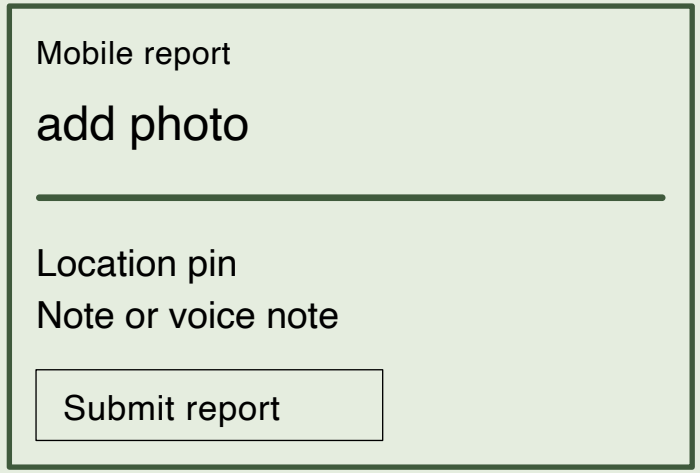


photo-first reporting

Photo, location pin, and voice note via a simple mobile interface.



shared visibility

Explore hazards by category, severity, location, and status.



municipal intelligence

Ranked queue, assignment, status tracking, repair verification.

Conceptual interface panels, not screenshots of a working product.

Planned citizen journey

from spotting a hazard to submitting a report in seconds

01 capture

Take a photo or upload existing evidence.

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02 pin

Confirm the location on the map.

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03 describe

Choose a category; add a note or voice note.

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04 submit

Get a report ID and track updates.

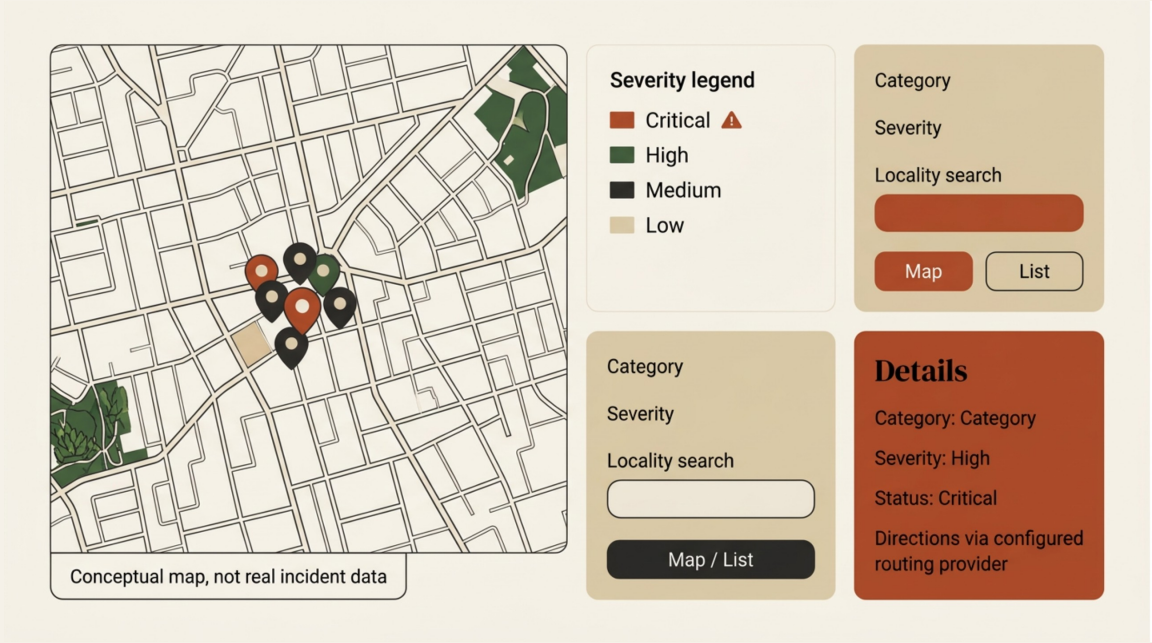
Designed for the moment someone notices a problem and needs the simplest way to report it.





Planned public map · conceptual design

turn scattered reports into a shared picture of street-level risk



Clustered locations, category and severity filters, locality search, incident details and status, map and list views, and directions through a configured routing provider.

A reported hazard is not a confirmed road closure, and routes are not called safe without verified information.

Planned municipal dashboard · conceptual design

help civic teams focus on what matters most

incident queue workspace

Queue control	Review context	Team action
Severity-ranked incidents	Explainable score	Assign municipal team
Time unresolved	Status history	Track progress
Repeat problems	Related reports	Review recurrence
Workload	Response analytics	Adjust priorities

Conceptual workspace structure; no live incident records or measured results.

explain the priority

Configurable severity scoring would consider hazard type, immediate danger, exposure, and corroborating evidence.

Authorised staff could make manual priority adjustments.

Ranking supports civic judgment; the rationale stays visible.

a report is only the beginning. proof of repair completes the loop.



Original evidence · illustrative photograph



After-repair evidence · illustrative photograph

01

report & act

Reported → Triaged → Assigned → In Progress

02

submit & review

Repair Submitted → Verification Pending

03

confirm closure

Verified Resolved. Rejected repairs reopen the incident.

Teams submit repair photos. Reviewers compare evidence. Authorised staff approve or reject. Citizens see the resulting status.

Close the loop with evidence, not just a status update. Images illustrate the concept, not an actual repair result.

Planned architecture · not yet built

a simple citizen experience. a robust system underneath.

experience layer

Responsive web app, photo capture, voice notes, interactive map.

application layer

Validation, report creation, severity scoring, assignment, notifications, verification.

data layer

PostgreSQL, geospatial data, secure media storage, authentication, audit history.

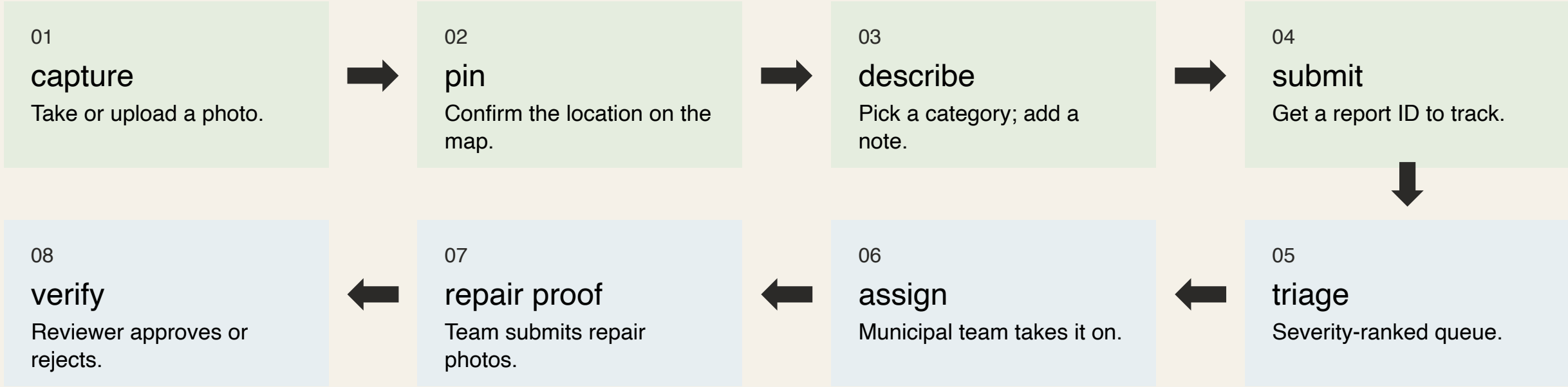
live services

Realtime updates, geocoding, map tiles, routing.

Planned workflow · conceptual

How the web app works: from first photo to verified repair

citizen · mobile web app



Verified resolved: citizens see the final status. Rejected repairs reopen the incident. Conceptual flow, not a built product.

Intended stack · not yet built

7 technologies, 4 layers, one planned stack

frontend · 2

React, TypeScript

maps · 1

MapLibre or Leaflet (one to be chosen)

backend & data · 3

Supabase, PostgreSQL, PostGIS

analytics · 1

Recharts

Count: React, TypeScript, MapLibre/Leaflet, Supabase, PostgreSQL, PostGIS, Recharts. Routing and geocoding providers not counted.